

Random Graph

Foundation of Data Science

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The $G(n, p)$ Model

The $G(n, p)$ Model

- The **random graph** $G(n, p)$ denotes the graph with n vertices and the probability that the edge is present is p .
- The presence of each edge is independent to each other.
- Interestingly, some global properties of the graph emerge, and there exists a **phase transition** according to value of p .

Degree Distribution

- The **degree** of vertex is the number of edges attached to it.
- Fix one vertex and let X be a random variable representing the degree of the vertex.
- Then, X follows the binomial distribution $B(n-1, p)$, i.e.,

$$P(X = k) = \binom{n-1}{k} p^k (1-p)^{n-1-k}.$$

- In fact, the degree distribution of the graph in real world does not follow a binomial distribution, but it resembles a power law distribution:

$$P(X = k) \propto \frac{1}{k^r}.$$

Existence of Triangles in $G(n, d/n)$

- For the vertices i, j, k , we define Δ_{ijk} as a indicator variable for the triangle with vertices i, j , and k being present, i.e.,

$$\Delta_{ijk} = \begin{cases} 1, & \text{if there is triangle with vertices } i, j, k, \\ 0, & \text{otherwise} \end{cases}.$$

- Define the total number of triangle as $x = \sum_{i,j,k} \Delta_{ijk}$. Then the expected value of x , which is the expected number of triangle is

$$E(x) = \sum_{i,j,k} E(\Delta_{ijk}) = \binom{n}{3} \left(\frac{d}{n}\right)^3 \approx \frac{d^3}{6}.$$

- However, this does not imply the existence of triangle in $G(n, d/n)$.
- To estimate $P(x = 0)$, we have the following geneous idea, together with Markov inequality:

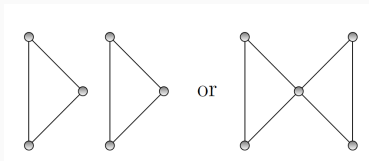
$$P(x = 0) \leq P(|x - E(x)| \geq E(x)) \leq \frac{V(x)}{E^2(x)} = \frac{E(x^2) - E^2(x)}{E^2(x)}.$$

Existence of Triangles in $G(n, d/n)$

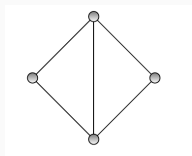
- However, we can estimate $E(x^2)$ as

$$E(x^2) = E \left(\left(\sum_{i,j,k} \Delta_{ijk} \right)^2 \right) = E \left(\sum_{i',j',k'} \Delta_{ijk} \Delta_{i'j'k'} \right).$$

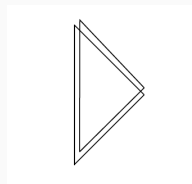
- We divide the set of i, j, k and i', j', k' into three groups, S_1, S_2 and S_3 .



(a) S_1 , edges are independent



(b) S_2 , share one edge



(c) S_3 , coincide

- Then, we estimate expected values on each subset as

$$E \left(\sum_{S_1} \Delta_{ijk} \Delta_{i'j'k'} \right) \leq E \left(\sum_{i,j,k} \Delta_{ijk} \right) E \left(\sum_{i',j',k'} \Delta_{i'j'k'} \right) = E(x)^2.$$

$$E \left(\sum_{S_2} \Delta_{ijk} \Delta_{i'j'k'} \right) = O(n^4 p^5) = O(d^5/n) = o(1).$$

$$E \left(\sum_{S_3} \Delta_{ijk} \Delta_{i'j'k'} \right) \leq E \left(\sum_{i,j,k} \Delta_{ijk}^2 \right) = E \left(\sum_{i,j,k} \Delta_{ijk} \right) = E(x) \approx \frac{d^3}{6}.$$

- Thus,

$$E(x^2) \leq E^2(x) + \frac{d^3}{6} + o(1),$$

and hence

$$P(x = 0) \leq \frac{d^3/6 + o(1)}{d^6/36} \leq \frac{6}{d^3} + o(1).$$

- Thus, if $d > 6^{1/3} \approx 1.8$, then $P(x = 0) < 1$ and thus, $G(n, p)$ has a triangle with nonzero probability.

Phase Transition

What is phase transition?

- In physics, the phase transition is a transition between states of matter, such as solid, liquid and gases. The phase transition occurs when the temperature passes the critical point.
- For the random graph $G(n, p)$, several properties of graph changes as the probability p passes “critical value”, or “threshold”.

Probability	Transition
$p = o(\frac{1}{n})$	Forest of trees, no component of size greater than $O(\log n)$
$p = \frac{d}{n}, d < 1$	Cycles appear, no component of size greater than $O(\log n)$
$p = \frac{d}{n}, d = 1$	Components of size $O(n^{\frac{2}{3}})$
$p = \frac{d}{n}, d > 1$	Giant component plus $O(\log n)$ components
$p = \frac{1}{2} \frac{\ln n}{n}$	Giant component plus isolated vertices
$p = \sqrt{\frac{2 \ln n}{n}}$	Diameter two
$p = \frac{\ln n}{n}$	Disappearance of isolated vertices Appearance of Hamilton circuit Diameter $O(\log n)$
$p = \frac{1}{2}$	Clique of size $(2 - \epsilon) \ln n$

Table 1: Phase transitions

Definition (Threshold)

1. For any property of graph, the function $p(n)$ is the **threshold** of that property if there is two functions $p_1(n)$ and $p_2(n)$ such that

$$\lim_{n \rightarrow \infty} \frac{p_1(n)}{p(n)} = 0, \quad G(n, p_1(n)) \text{ almost surely does not have property,}$$

$$\lim_{n \rightarrow \infty} \frac{p_2(n)}{p(n)} = \infty, \quad G(n, p_2(n)) \text{ almost surely has property.}$$

2. The function $p(n)$ is a **sharp threshold**, if

$G(n, cp(n))$ almost surely does not have the property for $c < 1$,

$G(n, cp(n))$ almost surely has the property for $c > 1$.

First moment and second moment method

- Let $x(n)$ be a number of occurrences of an item in a random graph. Then, from Markov inequality, we have $P(x \geq 1) \leq E(x)$. Thus, if $E(x(n)) \rightarrow 0$ as $n \rightarrow 0$, the probability of occurrence of the item decays to 0. This is called the **first moment method**.
- As we already observed in occurrence of triangle, if we have $V(x) = o(E^2(x))$, then x is almost surely greater than zero. This is called the **second moment method**.

Corollary

Let x be a random variable with $E(x) > 0$. If $E(x^2) \leq E^2(x)(1 + o(1))$, then x is almost surely greater than zero.

- Using these strategies, we can show that $p(n) = 1/n$ is a threshold of the property of having triangle.

Threshold for graph diameter two (two degrees of separation)

- The **diameter** of a graph is the maximum length of the shortest path between a pair of nodes. Thus, the graph with diameter two means the graph that each pair of vertices can be linked within two steps.

Theorem

The property that $G(n, p)$ has diameter two has a sharp threshold at

$$p = \sqrt{2} \sqrt{\frac{\log n}{n}}.$$

- We will call the pair of vertices (i, j) bad if they have distance greater than 2. We also define a indicator random variable I_{ij} with $i < j$, where $I_{ij} = 1$ if and only if (i, j) is bad. Finally, we define $x = \sum_{i < j} I_{ij}$, which is our main object.

Sketch of proof

- Then, we have

$$E(x) = \binom{n}{2} \underbrace{(1-p)}_{\text{not directly linked}} \times \underbrace{(1-p^2)^{n-2}}_{\text{not having common neighbor}}.$$

- Let $p = c\sqrt{\frac{\log n}{n}}$. Then, we have $E(x) \approx \frac{1}{2}n^{2-c^2}$ and thus, for $c > \sqrt{2}$,

$$\lim_{n \rightarrow \infty} E(x(n)) = 0.$$

Thus, by first moment method, $G(n, p)$ almost surely has no bad pair.

- On the other hand, for $c < \sqrt{2}$, by using similar argument as in counting triangle, we have

$$E(x^2) \leq E^2(x)(1 + o(1)),$$

which implies there are at least one bad pair almost surely for $G(n, p)$. This implies $p = \sqrt{2}\sqrt{\frac{\log n}{n}}$ is a sharp threshold.

Disappearance of Isolated Vertices

Theorem

The disappearance of isolated vertices in $G(n, p)$ has a sharp threshold of $\frac{\log n}{n}$.

- The idea of proof is similar. Let I_i be a indicator random variable that $I_i = 1$ if and only if i is an isolated vertex and define $x = \sum_i I_i$. Then

$$E(x) = n(1 - p)^{n-1}.$$

- For $p = c \frac{\log n}{n}$, we have

$$E(x) \approx n^{1-c}.$$

- Moreover, we have

$$\begin{aligned} E(x^2) &= \sum_{i=1}^n E(I_i^2) + 2 \sum_{i < j} E(I_i I_j) = \sum_{i=1}^n E(I_i) + n(n-1)E(I_1 I_2) \\ &= E(x) + n(n-1)(1-p)^{2(n-1)-1}. \end{aligned}$$

Disappearance of Isolated Vertices

- Therefore, for $p = \frac{c \log n}{n}$ with $c < 1$,

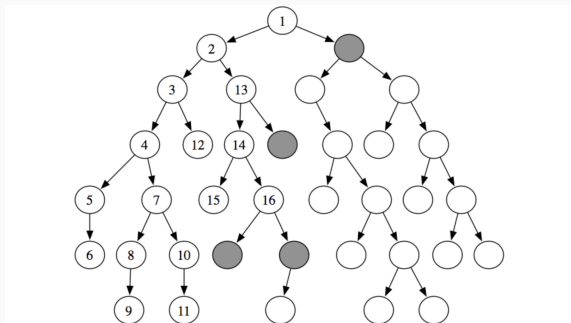
$$\begin{aligned}\frac{E(x^2)}{E^2(x)} &= \frac{1}{n(1-p)^{n-1}} + \left(1 - \frac{1}{n}\right) \frac{1}{1-p} \\ &\approx \frac{1}{n^{1-c}} + \left(1 - \frac{1}{n}\right) \frac{1}{1 - c \frac{\log n}{n}} = 1 + o(1).\end{aligned}$$

- Thus, by first and second moment methods, we can show that $p = \frac{\log n}{n}$ is a sharp threshold.

Giant Component

Existence of a giant component

- We consider $G(n, p)$ with $p = \frac{1+\varepsilon}{n}$. Then, we will show that with high probability, it has a giant component with size $\Omega(n)$. (Recall that $f(n) = \Omega(n)$ if there exists c such that $f(n) \geq cn$.)
- In order to show this, we do a depth first search (DFS) on $G(n, p)$ with $p = \frac{1+\varepsilon}{n}$, $0 < \varepsilon < \frac{1}{8}$.



Depth First Search (Source: https://artint.info/html/ArtInt_53.html)

Existence of a giant component

- We define the following subsets of vertices:

E :=Set of fully explored vertices whose exploration is complete.

U :=Set of unvisited vertices.

F :=Frontier of visited and still being explored vertices.

Lemma

After $\varepsilon n^2/2$ edge queries, with high probability $|E| < \frac{n}{3}$.

- Suppose not. Then, at some $t \leq \frac{\varepsilon n^2}{2}$, $|E| = \frac{n}{3}$. Moreover, the vertex is added to F if the query is answered yes. Thus, $|F| < \frac{\varepsilon n^2 p}{2} < \frac{n}{3}$ with high probability.
- Therefore, $|U| > n - \frac{2n}{3} = \frac{n}{3}$. Since E is the vertex whose exploration is completed, the queries between E and U should be answered as no, and the number of this query is at least $|E||U| > \frac{n^2}{9} > \frac{\varepsilon n^2}{2}$, which is contradiction.

Existence of a giant component

Lemma

After $\frac{\varepsilon n^2}{2}$ edge queries, with high probability the set F consists of at least $\frac{\varepsilon^2 n}{30}$ vertices.

- Suppose $|F| < \frac{\varepsilon^2 n}{30}$ after $\frac{\varepsilon n^2}{2}$ queries. Then,
 $|U| = n - |E| - |F| \geq n - \frac{n}{3} - \frac{\varepsilon^2 n}{30} > 1$. So, DFS is still alive. Then,
the expected number of yes answer so far is $\frac{p\varepsilon n^2}{2} = \frac{(1+\varepsilon)\varepsilon n}{2}$. Thus,
with high probability, the number of yes answer is at least $\frac{\varepsilon n}{2} + \frac{\varepsilon^2 n}{3}$.
- Thus, $|E| + |F| > \frac{\varepsilon n}{2} + \frac{\varepsilon^2 n}{3}$ and hence, $|E| > \frac{\varepsilon n}{2} + \frac{3\varepsilon^2 n}{10}$.
- Therefore,

$$\begin{aligned} |E||U| &= |E|(n - |E| - |F|) > \left(\frac{\varepsilon n}{2} + \frac{3\varepsilon^2 n}{10}\right) \left(n - \frac{\varepsilon n}{2} - \frac{3\varepsilon^2 n}{10} - \frac{\varepsilon^2 n}{30}\right) \\ &> \frac{\varepsilon n^2}{2}, \end{aligned}$$

which is contradiction.

- $|F| = \Omega(n)$ implies there exists a giant component.

Cycles and Full Connectivity

Emergence of cycles

Theorem

The threshold for the existence of cycles in $G(n, p)$ is $p = \frac{1}{n}$. This threshold is not sharp.

- Let x be the number of cycles in $G(n, p)$. Then,

$$\begin{aligned} E(x) &= \sum_{k=3}^n \binom{n}{k} \frac{(k-1)!}{2} p^k \leq \sum_{k=3}^n \frac{n^k}{2k} p^k \leq \sum_{k=3}^n (np)^k \\ &= (np)^3 \frac{1 - (np)^{n-2}}{1 - np} \leq 2(np)^3, \quad \text{when } np < \frac{1}{2}. \end{aligned}$$

- Thus, if p is asymptotically less than $\frac{1}{n}$, then, $E(x) \rightarrow 0$ as $n \rightarrow \infty$ and by first moment method, there is no cycle almost surely.
- For $p = \frac{d}{n}$ with $d > 1$, a graph will have a cycle with probability tending to 1, by second moment method.
- When $d < 1$, $E(x)$ converges to constant greater than 0.

Full connectivity

Lemma

The expected number of connected components of size k in $G(n, p)$ is at most $\binom{n}{k} k^{k-2} p^{k-1} (1-p)^{kn-k^2}$.

- The probability of k vertices form a connected component is (Existence of spanning tree within k vertices) $\times$ (Probability of no other edges between k vertices and others).
- There are k^{k-2} possible spanning trees within k nodes. Thus, the first probability is equal to $k^{k-2} p^{k-1}$ and the second probability is $(1-p)^{k(n-k)}$.

Theorem

For $p = c \frac{\log n}{n}$, with $c > \frac{1}{2}$, almost surely there are only isolated vertices and a giant component. For $c > 1$, almost surely the graph is connected.

- After we have the first assertion, the second assertion directly obtained, since we already shown that for $c > 1$, there is no isolated point.

- Let x_k be the number of connected components of size k . The goal is to show $E \left(\sum_{k=2}^{n/2} x_k \right) \leq n^{1-2c}$, hence when $c > \frac{1}{2}$, we may use first moment method. This can be obtained by using Lemma, with $p = c \frac{\log n}{n}$.

Phase Transition for Increasing Properties

Increasing property

- The property Q is said an **increasing property** if when a graph G has the property, any graph obtained by adding edges to G must also have the property.
- For example, connectivity, no isolated vertices, having cycle.

Lemma

If Q is an increasing property of graphs and $0 \leq p \leq q \leq 1$. Then, the probability that $G(n, q)$ has Q is greater than the probability that $G(n, p)$ has Q .

- An **m -fold replicant of $G(n, p)$** is a random graph that overlap all the edges of m copies of $G(n, p)$. This graph is nothing but $G(n, q)$ with $q = 1 - (1 - p)^m \leq mp$. Thus,

$$P(G(n, mp) \text{ has } Q) \geq P(G(n, q) \text{ has } Q).$$

Phase transition for increasing properties

Theorem

Each increasing property Q of $G(n, p)$ has a phase transition at $p(n)$, where $p(n)$ is the minimum real number a_n for which the probability that $G(n, a_n)$ has property Q is $\frac{1}{2}$.

- Let $p_0(n)$ be any function such that

$$\lim_{n \rightarrow \infty} \frac{p_0(n)}{p(n)} = 0.$$

Suppose the probability that $G(n, p_0)$ has Q does not converge to 0. Then, there exists ε such that $P(G(n, p_0) \text{ has } Q) > \varepsilon$ for infinitely many n . Let $m = \lceil \frac{1}{\varepsilon} \rceil$. Consider m -fold replication of $G(n, p_0)$, which is $G(n, q)$. Then,

$P(G(n, q) \text{ does not have } Q) \leq (1 - \varepsilon)^m \leq e^{-1} \leq \frac{1}{2}$. Thus,

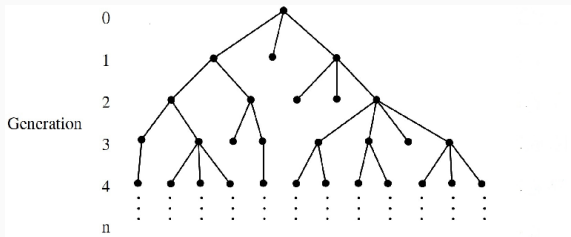
$$P(G(n, mp_0) \text{ has } Q) \geq P(G(n, q) \text{ has } Q) \geq \frac{1}{2},$$

and by definition of $p(n)$, we have $mp_0 \geq p(n)$, which is contradict to the definition of p_0 .

Branching Processes

Branching Process

- A **branching process** is a method for creating a random tree.
- Each node has a (same) probability distribution for the number of its children.



Branching process

(Source: Cao, Yong Xian et al. INFERENCE FOR AGE-DEPENDENT BRANCHING PROCESS AND THEIR APPLICATIONS. (2015).)

Branching process

- One possible question is “what is the probability that the branching process dies out?”
- In other words, what is the **extinction probability**?
- The **generating function** for a nonnegative integer-valued random variable y is $f(x) = \sum_{i=0}^{\infty} p_i x^i$, where $p_i = P(y = i)$.
- Let z_j be the number of children in the j -th generation and $f_j(x)$ be the generating function for z_j . Let $f(x) = f_1(x)$. By the definition, $f(x)$ is a generating function of the number of children for each node.
- We will show $f_{j+1}(x) = f_j(f(x))$. This follows from the following two facts:
 - The generating function of sum of two independent random variables is multiplication of two generating functions.
 - The coefficient of x^i in $f_j(x)$ is the probability of there being i children in the j -th generation.

- Thus, we have

$$f_{j+1}(x) = \sum_{i=0}^{\infty} P(z_j = i) f^i(x) = f_j(f(x)).$$

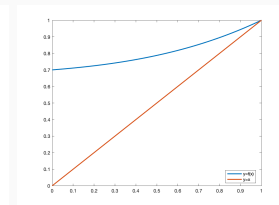
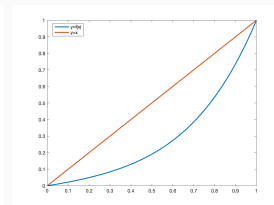
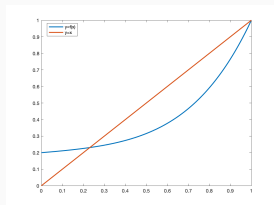
- Now, let q be a extinction probability. Suppose there exists i nodes in the first generation. Then, we have

$$q = \sum_{i=0}^{\infty} p_i q^i = f(q).$$

- Thus, q is a root of $x = f(x)$. Since p_i are nonnegative, $f(x)$ is an increasing convex function.
- Let $m = f'(1) = \sum_{i=1}^{\infty} i p_i$, which is the expected number of children for each node.

Extinction Probability

- Depending on m and p_0 , we have different asymptotic behaviors for branching process.



(a) $m > 1$, $p_0 > 0$, $0 < q < 1$

(b) $m > 1$, $p_0 = 0$, $q = 0$

(c) $m < 1$, $q = 1$

- We can interpret the result as follows: If the expected number of children (m) is larger than 1 and if the probability of die out in one generation (p_0) is nonzero, then $0 < q < 1$. If $p_0 = 0$ then, $q = 0$, which implies the extinction probability is 0. When $m < 1$, process dies out almost surely.

**THANK YOU VERY MUCH FOR YOUR
ATTENTION.**